Making every real estate decision faster and smarter

DATATHON 2026

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved

robinreal

The Challenge

User queries mix hard constraints with soft preferences

Your system needs to separate these two layers and combine them effectively.

QUERY 1

"**3-room** *bright* apartment in **Zurich under 2800 CHF** with **balcony**, *close to public transport*"

■ Hard ■ Soft

QUERY 2

"*Bright family-friendly* flat in **Winterthur**, *not too expensive*, *ideally with parking*"

Mostly soft preferences — harder to rank

QUERY 3

"*Modern* **studio** in **Geneva** for **June move-in**, *quiet area*, *nice views if possible*"

Conflicting: cheap vs. central, quiet vs. views

■ **Hard constraints** = must be respected ■ **Soft preferences** = should influence ranking

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved

robinreal

Your Task

Map a natural-language query to a ranked list of listings

STEP 1
Extract Hard Filters
Parse the query into strict criteria: rooms, price, city, features

→

STEP 2
Retrieve Candidates
Query listings that satisfy all hard constraints via the API

→

STEP 3
Rank by Relevance
Score soft preferences and sort by overall relevance

→

OUTPUT
Ranked Results
Sorted listing IDs with scores + explanations

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved
robinreal

The Data

Listing Data Structure

IDENTITY & PROPERTY

listing_id 36272162
title "3.5 Zimmer Wohnung"
description full text
category "Wohnung"
marketing_type RENT
status ACTIVE / INACTIVE

LOCATION & GEO

street "38 Röhrliberg"
city "Cham"
postal_code "6330"
canton "ZG"
latitude / longitude 47.186 / 8.451

PRICING & SIZE

price 2450 (CHF/month)
num_rooms 3.5
living_area 97 m²
floor 2
available_date 2025-12-12
construction_year 1979

FEATURES (boolean flags)

HasBalconies ✓ HasBaths ✓ HasParkingIndoor ✓ HasWashingmachine ✓
HasElevator HasGarden HasPool HasFireplace PetsAllowed …

Structured flags where available — not all listings have complete feature data

MEDIA

image_urls array of CDN links
local image bundle

Images useful for multimodal ranking

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved
robinreal

Getting Started

Your Starter Kit

https://shorturl.at/NRPAw

HARD FILTER API

Pre-built filtering service. Extract constraints from queries and filter listings.

rooms, price, city, features

AI SDK APP

Scaffolded app with Claude **(provided by Datathon)** for LLM-powered query parsing.

Claude 4 Sonnet ready to go

LISTING DATASET

Swiss rental listings hosted on AWS **(provided by Datathon)**. Structured data + images.

~thousands of listings on S3

IMAGE BUNDLE

Property photos for multimodal analysis. Room detection, quality scoring, visual ranking.

jpg, multiple per listing

DOCKER & COMPOSE

Full local dev environment. One command to spin up all services.

docker compose up

MCP SERVER

Model Context Protocol server for tool-augmented AI agent workflows.

connect your AI to live data

💡 We recommend using **AWS** and **Claude** provided by Datathon — credits are included for all participants.

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved

robinreal

DATA ENRICHMENT

Go Beyond the Dataset

Listings have basic data — enrich them with external sources to unlock stronger ranking signals.

Transport & Commute

- SBB API for travel times
- Public transport stops nearby
- Commute duration to key hubs
- Proximity to train stations

Amenities & Services

- Supermarkets, pharmacies, schools
- Restaurants & nightlife density
- Healthcare facilities nearby
- OpenStreetMap POI data (Nominatim)

Geospatial Features

- Geocoding via Nominatim
- Neighborhood classification
- Walkability & bike scores
- Elevation & noise levels

Neighborhood Context

- Average rents in the area
- Safety & crime statistics
- Demographic profiles
- Green space & parks ratio

Image Analysis

- Room brightness & natural light
- Modernity of interiors
- View quality from windows
- Space perception & layout

Text Enrichment

- Extract features from descriptions
- Sentiment of listing language
- Keyword tagging (cozy, modern)
- Multilingual normalization

EXAMPLE — WHY ENRICHMENT MATTERS

"I want a quiet apartment **near ETH** with a **short commute to the city center**, **modern-looking** inside, with **good schools nearby**"

→ **near ETH** needs Geospatial **short commute** needs Transport **modern-looking** needs Image Analysis **good schools** needs Amenities

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved

robinreal

Example 1: Clear Hard Constraints

User Query

"**3.5-room** bright apartment in **Zurich** under **CHF 2800** with **balcony**"

Bold = hard filter Orange = soft preference

Candidate Listings After Hard Filters (city = Zürich, rooms = 3.5, price ≤ 2800, balcony = true)

Helle 3.5-Zi an der Wengistrasse

Wengistrasse 2, 8004 Zürich

CHF 2,650/mo

3.5 rooms · 85 m² · 2nd floor

Balcony ✓ Minergie ✓ Washer ✓

Bright Minergie-certified apartment in Kreis 4. Large south-facing windows, modern kitchen, parquet floors.

⭐ **bright + Minergie + central**

Moderne Wohnung Oerlikon

Käferholzstrasse 42, 8057 Zürich

CHF 2,450/mo

3.5 rooms · 75 m² · 3rd floor

Balcony ✓ Washer ✓ Parkett ✓

Maisonette near Bucheggplatz. Rooftop terrace, close to ETH Hönggerberg campus. Well connected.

🏠 **maisonette + terrace**

Gemütliche Wohnung Seebach

Hertensteinstrasse 8, 8052 Zürich

CHF 2,200/mo

3.5 rooms · 78 m² · 4th floor

Balcony ✓ Washer ✓ Parking ✓

Private terrace with access from all rooms. In-unit washer/dryer. Near Seebach station. Quiet area.

🌳 **quiet + parking + value**

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved

robinreal

Example 1: Ranking Output

How should these rank for "bright"?

#1

Helle 3.5-Zi, Wengistrasse

CHF 2,650 score: 0.94

Why this rank?

- "Helle" (bright) in title — strong signal
- Minergie = large windows, energy-efficient glass
- South-facing, 2nd floor — maximum light
- Modern 2019 build with open floor plan

#2

Moderne Wohnung, Oerlikon

CHF 2,450 score: 0.82

Why this rank?

- Maisonette = windows on two levels
- Roof terrace — skylights likely
- Near ETH campus — well-maintained area
- No explicit “bright” signal in listing

#3

Gemütliche Wohnung, Seebach

CHF 2,200 score: 0.71

Why this rank?

- “Gemütlich” (cozy) ≠ bright
- 4th floor = good light potential
- Best price, but weaker brightness signal
- Parking + terrace still strong amenities

Key insight: "Bright" is a soft preference — your ranking system must interpret description text + metadata to score it

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved

robinreal

Example 2: Vague & Soft-Heavy

User Query

"Bright family-friendly flat in **Winterthur**, not too expensive, ideally with parking"

Challenge: Only "Winterthur" is a hard filter. Everything else is subjective. How does the system interpret "not too expensive"?

System Must Interpret

- **"not too expensive"** → below Winterthur median? bottom 30%? Infer from context: 3.5 rooms → ~CHF 1,600–2,450 range
- **"family-friendly"** → 3+ rooms, near schools/parks, quiet area, ground floor or elevator, pets allowed?
- **"bright"** → large windows, high floor, south-facing, or "hell"/"lichtdurchflutet" in description
- **"ideally with parking"** → bonus, not a hard requirement. HasParkingIndoor flag, or mention in description

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved
robinreal

Example 3: Conflicting Preferences

User Query

"Modern apartment in **Basel**, cheap but central, quiet with nice views"

Conflict: "Cheap" and "central" are inversely correlated. "Quiet" and "central" often conflict. How does your system balance tradeoffs?

Strategy A

Prioritize Price

2.5-Zi Kleinbasel, Klybeckstr.

CHF 1,350/mo

48 m² · 2nd floor · 1995 build

✓ Cheapest option
✓ Central Kleinbasel location
✗ Street noise (Klybeck is busy)
✗ Older, not “modern”
✗ No views to speak of

Strategy B

Balance All Signals

3.5-Zi Gundeldingen, Güterstr.

CHF 1,950/mo

72 m² · 5th floor · 2021 build

✓ Modern new build
✓ 5th floor = views + quiet
✓ 10 min to SBB by tram
✗ Not the cheapest
✗ “Central” is debatable

Strategy C

Prioritize Lifestyle

3.5-Zi St. Alban, Münsterplatz

CHF 2,800/mo

85 m² · 3rd floor · Renovated

✓ Premium central location
✓ Rhine views, historic quiet
✓ High-end renovation
✗ Not cheap at all
✗ May exceed user budget

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved

robinreal

EXAMPLE — IMAGE ANALYSIS FOR RANKING

What Can You Learn from Photos?

Vision models can extract soft ranking signals that text alone misses

☀️ Brightness & Light

Detect natural light levels, window size, south-facing rooms, time-of-day photography bias

✨ Interior Modernity

Classify kitchen/bathroom age, identify recent renovations, detect style (modern vs. traditional)

📐 Space & Layout

Estimate perceived spaciousness, detect open floor plans, measure visual clutter

🏞️ Views & Surroundings

Identify window views (greenery, cityscape, mountain), balcony presence, outdoor spaces

How to use image signals

Combine vision scores with text-based relevance for multimodal ranking. A listing described as "bright" that also shows dark photos should be penalized.

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved

robinreal

Approaches

Solution Directions

LLM QUERY DECOMPOSITION

Use GPT-4o to parse natural language into structured filters + soft preferences. Handle ambiguity, infer missing constraints.

💡 *Prompt engineering is key*

HYBRID SEARCH

Combine keyword search (BM25) with semantic embeddings. Structured filters narrow candidates, embeddings rank by meaning.

💡 *Best of both worlds*

EMBEDDING-BASED RANKING

Embed queries and listing descriptions into shared vector space. Rank by cosine similarity. Fine-tune for real estate domain.

💡 *Handles soft preferences well*

MULTIMODAL SIGNALS

Use listing images to assess brightness, modernity, views. Vision models can extract features invisible in structured data.

💡 *Images reveal what text hides*

GEOSPATIAL ENRICHMENT

Use lat/lng to compute distances to transit, schools, parks. Enrich listings with neighborhood characteristics.

💡 *Great for "central" or "quiet"*

LEARN-TO-RANK

Train a model on pairwise preferences. Combine all signals (text, features, geo, images) into a unified relevance score.

💡 *Advanced but powerful*

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved

robinreal

BONUS

User Preference History

Personalize ranking using past interaction signals. This is optional but will impress the jury.

Interaction Signals

- **Favorites / saves** — explicit positive signal
- **Click-through rate** — which listings get opened
- **Dwell time** — time spent on a listing page
- **Query refinement patterns** — how users narrow down
- **Dismissals / skips** — implicit negative signal

Build a user profile that shifts ranking weights over time. A user who always saves modern apartments should see them ranked higher.

Example Scenario

A user has previously:

- Saved 5 apartments with Minergie label
- Spent 3x more time on listings with balcony photos
- Skipped all listings above CHF 2,500
- Always searched in Kreis 4–6

→ **Inferred profile:**

Eco-conscious, outdoor living, budget-sensitive, urban core preference. Boost Minergie + balcony + central + affordable in ranking weights.

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved

robinreal

Judging

Evaluation & Judging

Peer Review – Hidden and public test set

- **Peer feedback** — Other teams evaluate your output
- **Prepared Public queries**
- **Hidden Test queries**
- **Hard-Filter Precision** — Do results actually satisfy hard constraints?

Jury & Peer Review

- **Technical depth** — Quality of architecture and approach
- **Feature width** — Variety of different approaches and considerations
- **Creativity** — Novel ideas, unexpected data usage
- **Demo quality** — Working prototype, clear presentation
- **Failure analysis** — Show where your system breaks
- **Peer feedback** — Other teams evaluate your output

Deliverables

① Working prototype ② Technical write-up ③ Live demo ④ Ranking logic explanation

Submission: Live GitHub repo with running code & presentation

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved

robinreal

Job Opportunity

Looking for a world-class AI Engineer to join the mission.

AI-Engineer

- Start: asap
- Workload: 40-100%
- Salary: 30-45 CHF/h
- Location: Zurich Accelerator Office

Applications: info@robinreal.ai

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved

robinreal

Let’s hack!

DATATHON 2026

philipp@robinreal.ai
Tel: +41 76 262 33 21

Confidential · robinreal® (SwissBid GmbH) · © 2026 All rights reserved

robinreal